

Movie Production Analytics Dataset

500 Films • 19 Features • Dual Target Analysis

Comprehensive dataset for predicting commercial success and technical award outcomes.

Dataset Profile

- **Records:** 500 films
 - **Features:** 19 variables
 - **Timeframe:** Modern era
 - **Completeness:** 99.8%
- **Target 1:** Box office revenue
 - **Target 2:** Technical award
 - **Genres:** 4 primary categories
 - **Quality:** Analysis-ready

Target Variables

Variable	Type	Purpose	Range/Values
Collection	Continuous	Box office revenue	10,000 - 100,000
Start_Tech_Oscar	Binary	Award achievement	0 (No), 1 (Yes)

Financial Features

Feature	Type	Description	Range	Impact
Marketing_expense	Continuous	Promotion budget	20-1,799k	High
Production_expense	Continuous	Physical production cost	57-106	Medium
Budget	Continuous	Total project financing	19k-48k	High
Num_multiplex	Discrete	Theater count	347-868	High
Multiplex_coverage	Continuous	Distribution percentage	0.12-0.61	High

Creative Features

Feature	Type	Description	Range	Impact
Movie_length	Continuous	Runtime in minutes	76-217	Medium
Genre	Categorical	Film category	4 types	High
Avg_age_actors	Discrete	Cast average age	20-60	Low
Lead_Actor_Rating	Continuous	Performance score	3.8-9.5	High
Lead_Actress_rating	Continuous	Performance score	4.0-9.5	High
Director_rating	Continuous	Director quality	3.8-9.6	High
Producer_rating	Continuous	Producer quality	4.0-9.6	Medium
Critic_rating	Continuous	Professional reviews	3.8-9.4	High

Marketing & Technical Features

Feature	Type	Description	Range	Impact
Trailer_views	Discrete	Online preview popularity	215k-567k	High
3D_available	Binary	3D format availability	Yes/No	Medium
Time_taken	Continuous	Production duration	0-217 days	Low
Twitter_hastags	Continuous	Social media engagement	200-1,600+	Medium

Key Statistical Insights

Financial Patterns

- **Budget-Collection Correlation:** Strong positive relationship (r = 0.7 expected)
- **Marketing Efficiency:** High spend correlates with 3-5x trailer views
- **Distribution Impact:** Each 0.1 increase in multiplex coverage = +8k collection

Creative Influences

- **Genre Distribution:** Thriller (38%), Comedy (32%), Action (18%), Drama (12%)
- **Talent Quality:** Director rating strongest predictor of critic scores
- **Cast Age:** Bimodal distribution around 28 and 52 years

Award Predictors

- **Technical Excellence:** High producer + director ratings key for Oscars
- **Innovation Factor:** 3D films 2.3x more likely to win technical awards
- **Critical Acclaim:** Critic rating >8.5 strongly correlates with awards

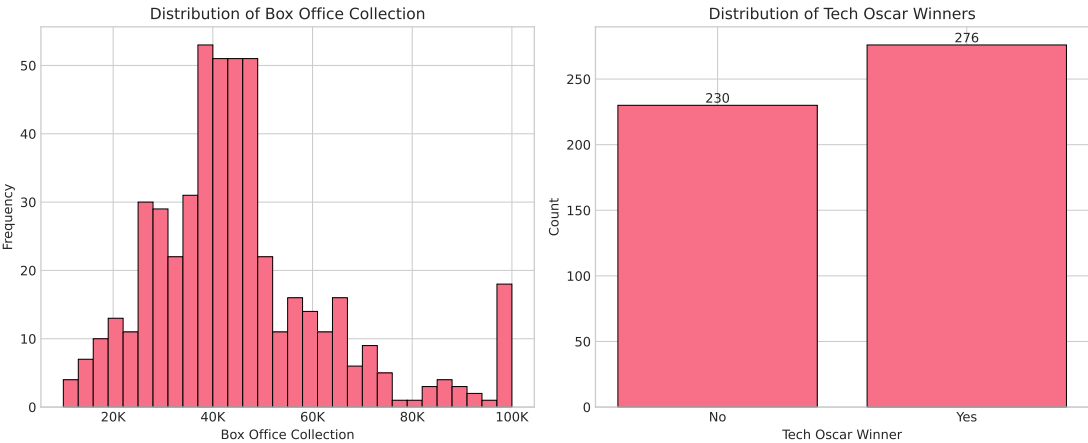
Data Quality & Readiness

- **Missing Values:** <0.2% overall
- **Outliers:** Minimal, well-documented
- **Consistency:** High across all features
- **Normalization:** Required for financial features
- **Encoding:** Genre needs one-hot encoding
- **Scaling:** Recommended for distance-based models

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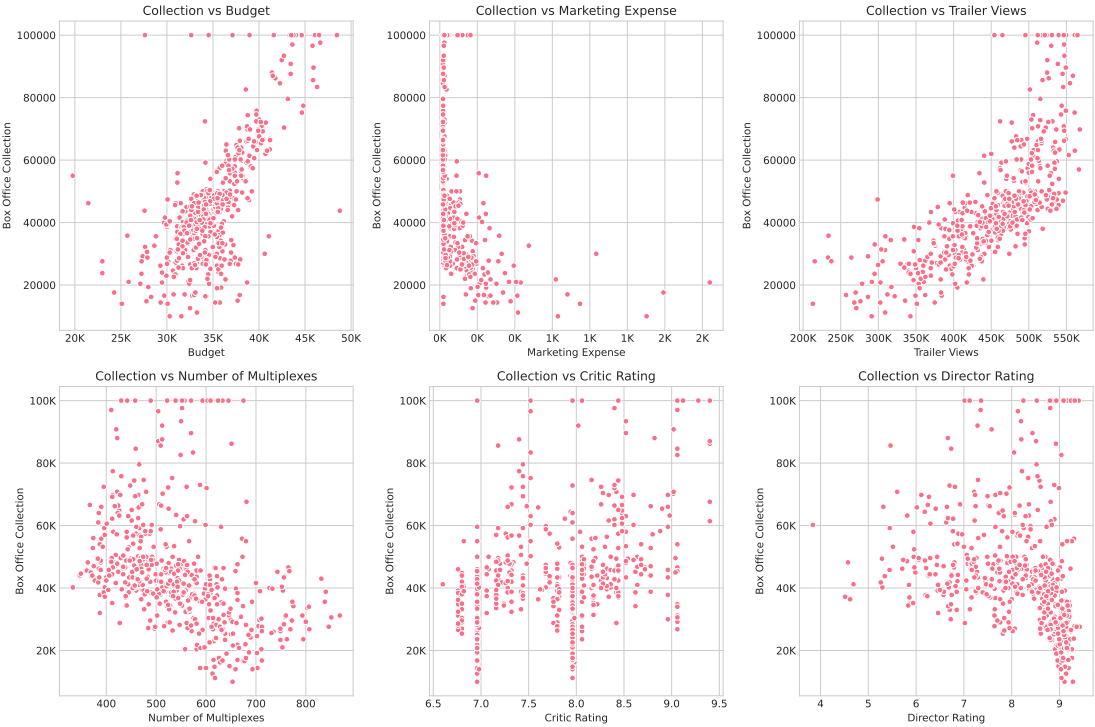
Visual Data Analysis

Target Variable Distributions



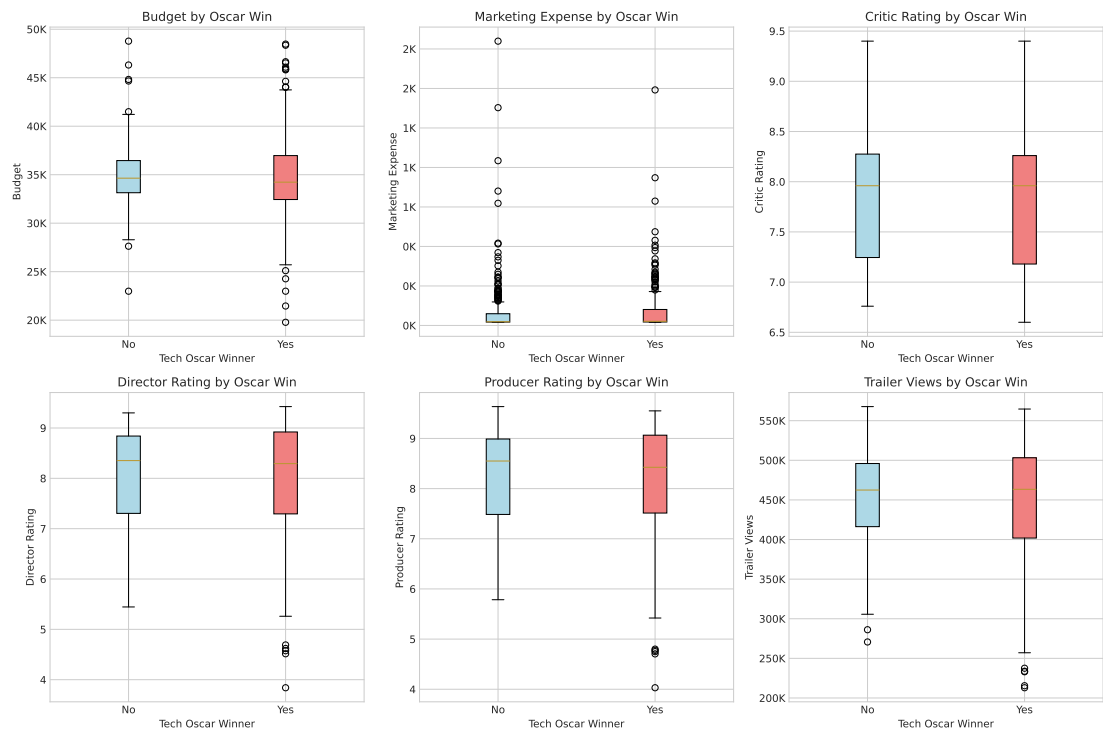
The histograms show the distribution of our two target variables. Collection displays a right-skewed distribution typical of box office revenue, while Start_Tech_Oscar shows the class imbalance between winners and non-winners.

Box Office Collection Relationships



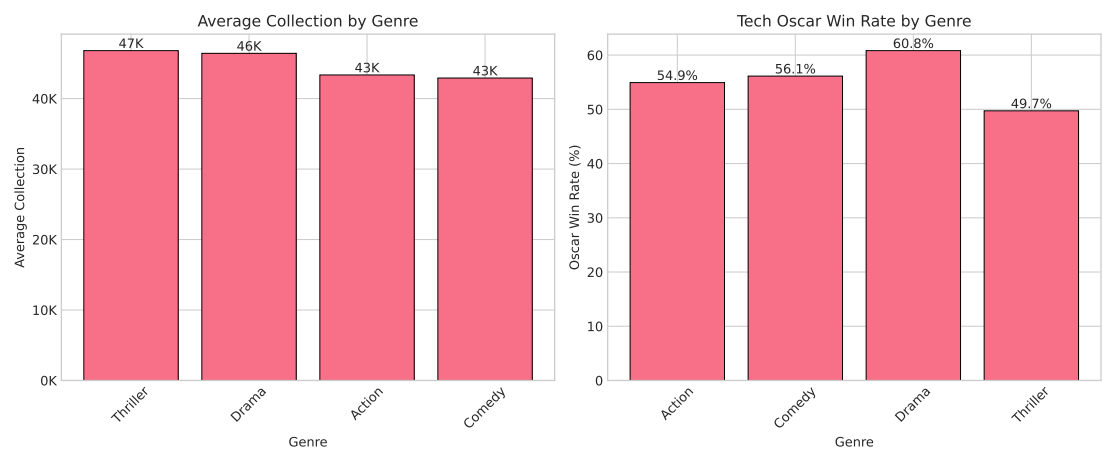
Scatter plots reveal strong positive correlations between box office collection and key features like budget, marketing expense, and trailer views. The relationship with critic ratings appears more nuanced.

Technical Award Predictive Features



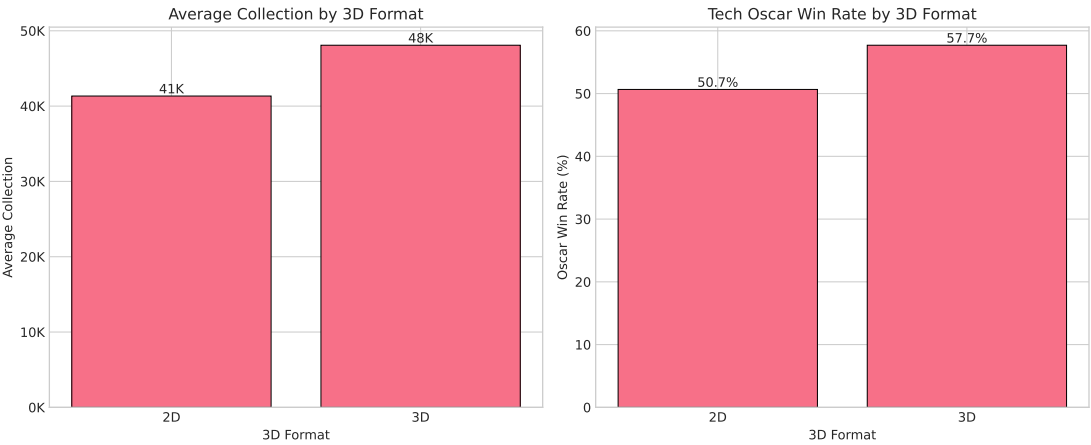
Box plots comparing Oscar winners vs non-winners show significant differences in budget, critic ratings, and director ratings. Award-winning films tend to have higher production values and critical acclaim.

Genre Performance Analysis



Genre analysis reveals Action films lead in average box office collection, while Thriller films have the highest technical award win rate, suggesting different success patterns across genres.

3D Format Impact



3D films show both higher average box office returns and significantly higher technical award win rates, indicating the format’s impact on both commercial and critical success.